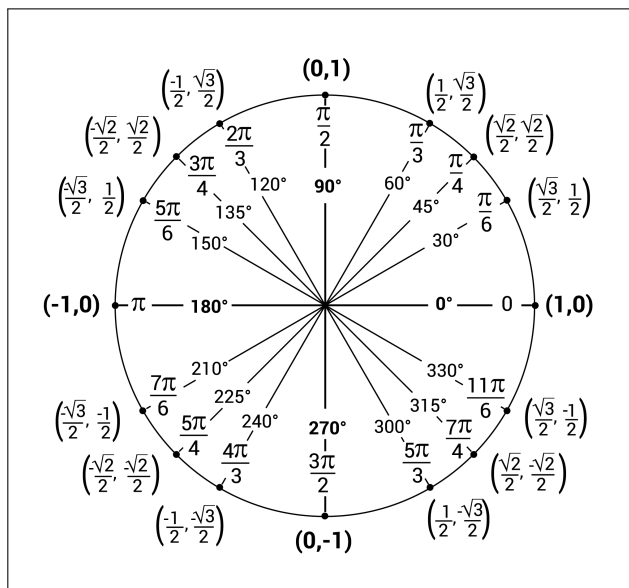


Extra Unit B practice 2

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the final practice worksheets, and the practice problems. The formulas below will be provided on Opportunity B.

Formulas

- $L(x) = f'(a)(x - a) + f(a)$
- MVT: $\frac{f(b) - f(a)}{b - a} = f'(c)$
- Basic derivatives
 - $\frac{d}{dx}(x^n) = nx^{n-1}$
 - $\frac{d}{dx}(e^x) = e^x$
 - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx}(\sin x) = \cos x$
 - $\frac{d}{dx}(\cos x) = -\sin x$
 - $\frac{d}{dx}(\tan x) = \sec^2 x$
 - $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx}(\cot x) = -\csc^2 x$
 - $\frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
 - $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$



B1 - Extrema of functions

(a) A function and its derivative are given below.

$$f(x) = \frac{x+1}{x^2+3x+3}$$

$$f'(x) = \frac{-x(x+2)}{(x^2+3x+3)^2}$$

Find the absolute maximum and absolute minimum of $f(x)$ on the interval $[-9, -1]$. Report your answers as ordered pairs.

not in $[-9, -1]$
 $f'(x) = 0$ @ $x=0$ and $x=-2$

$$f(-9) = \frac{-9+1}{(-9)^2+3(-9)+3} = \frac{-8}{81-27+3} = \frac{-8}{57}$$

$$f(-2) = \frac{-2+1}{(-2)^2+3(-2)+3} = \frac{-1}{4-6+3} = -1 \leftarrow \text{abs. min at } (-2, -1)$$

$$f(-1) = \frac{-1+1}{(-1)^2+3(-1)+3} = 0 \leftarrow \text{abs max at } (-1, 0)$$

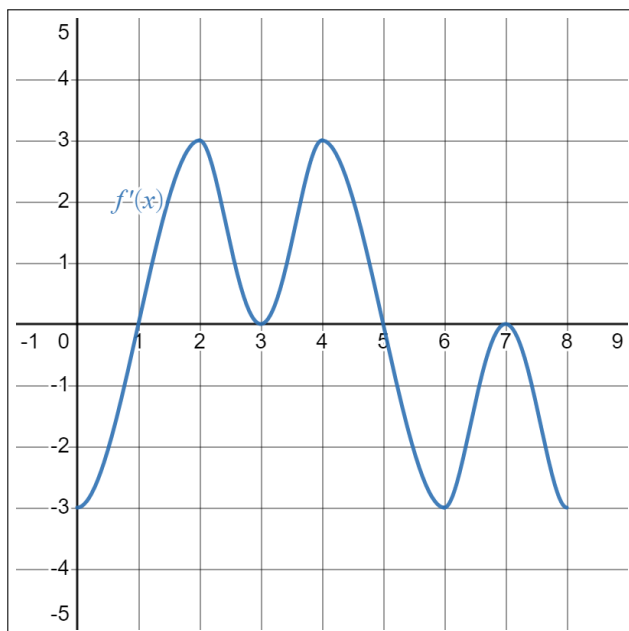
(b) The graph of the derivative $f'(x)$ of a function $f(x)$ is shown to the right.

(i) At what value(s) of x does $f(x)$ have a local maximum?

$$x = 5$$

(ii) On what interval(s) is $f(x)$ concave down?

$$[2, 3], [4, 6], [7, 8]$$



B2 - The Mean Value Theorem

(a) Show that the equation $f(x) = 3x + 2\cos x + 5 = 0$ has exactly one real root.

$f(x)$ is continuous (linear + trig), so we can use IVT.

$$\begin{aligned} f(-\pi) &= 3 \cdot (-\pi) + 2 \cdot \cos(-\pi) + 5 = -3\pi + 3 < 0 \\ f(0) &= 3 \cdot 0 + 2 \cos(0) + 5 = 7 > 0 \end{aligned} \quad \Rightarrow \quad \text{so } f(x) \text{ has at least 1 real root in } (-\pi, 0).$$

$f(x)$ is differentiable, so we can use MVT.

$$f'(x) = 3 - 2\sin x = 0 \rightarrow \sin x = \frac{3}{2} \rightarrow \text{no solutions}$$

Since $f'(x) = 0$ has no solutions, by MVT $f(x) = 0$ has at most 1 solution.

$\rightarrow$ at least 1 and at most 1 $\rightarrow f(x) = 0$ has exactly 1 solution.

(b) By applying the Mean Value Theorem to the function $f(x) = x^{1/5}$ on the interval $[32, 33]$, show that

$$\sqrt[5]{33} < 2 + \frac{1}{5}(32)^{-\frac{4}{5}}$$

Hint: The derivative of $x^{\frac{1}{5}}$ is a decreasing function, so its largest value on the interval $[32, 33]$ occurs at $x = 32$.

$$\frac{f(33) - f(32)}{33 - 32} = f'(c) < f'(32)$$

$$f'(x) = \frac{1}{5}x^{-\frac{4}{5}}$$

$$\frac{\sqrt[5]{33} - \sqrt[5]{32}}{1} < \frac{1}{5}(32)^{-\frac{4}{5}}$$

$$\sqrt[5]{33} - 2 < \frac{1}{5}(32)^{-\frac{4}{5}} \rightarrow \sqrt[5]{33} < 2 + \frac{1}{5}(32)^{-\frac{4}{5}}$$

B3 - Curve sketching

Consider the function:

$$f(x) = \frac{x}{x^2 - 1}$$

The first and second derivatives of this function are given by the formulas:

$$f'(x) = \frac{-x^2 - 1}{(x^2 - 1)^2} \qquad f''(x) = \frac{2x^3 + 6x}{(x^2 - 1)^3}$$

In all of the following questions, you must use calculus (not a graphing calculator) to justify your answers.

- (a) What are the horizontal and vertical asymptotes of f (if any)?

$$x=1, \quad x=-1, \quad y=0$$

- (b) On which interval(s) is f increasing?

None

- (c) What are the x -values of the local maxima/minima of f (if any)?

None

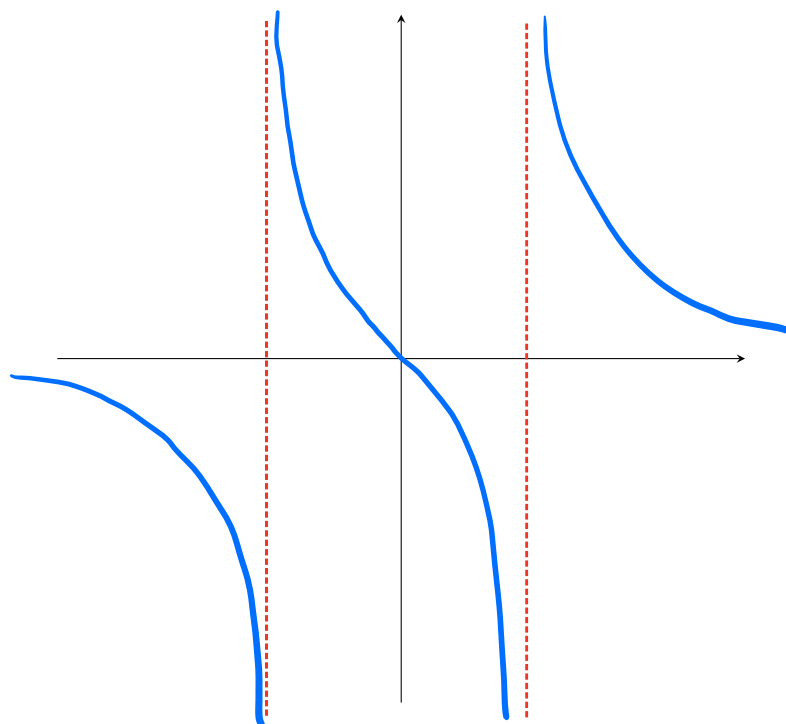
- (d) On which interval(s) is f concave down?

$$(-\infty, -1) \quad (0, 1)$$

- (e) What are the x -values of the points of inflection of f (if any)?

$$x=0$$

- (f) Sketch a graph of f below and label the x -intercepts, asymptotes, relative extrema, and points of inflection.

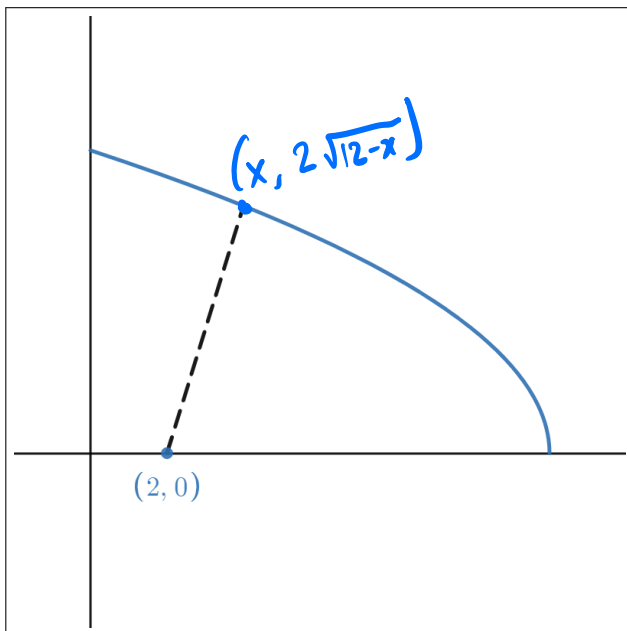


B4 - Optimization

The graphs below, each showing $f(x) = 2\sqrt{12-x}$ for $x > 0$, are provided to help you solve the problems.

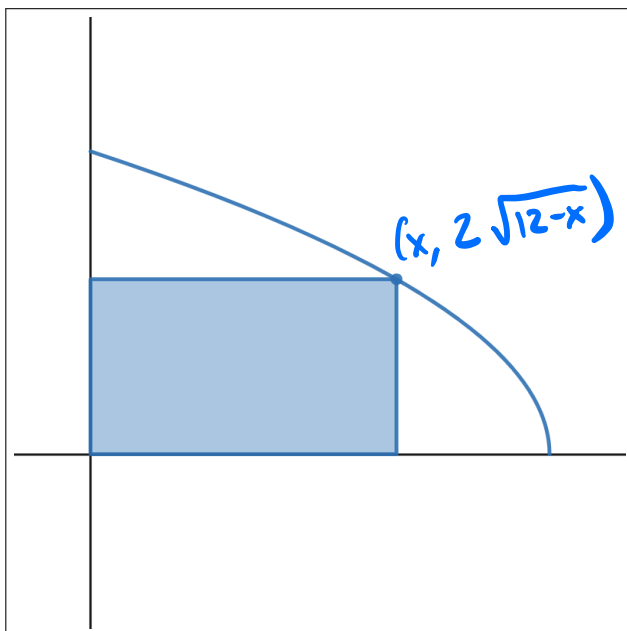
- (a) What is the shortest distance between the point $(2, 0)$ and a point on the graph of $f(x)$?

$$\begin{aligned} D^2(x) &= (x-2)^2 + (0-2\sqrt{12-x})^2 \\ &= x^2 - 4x + 4 + 4 \cdot (12-x) \\ &= x^2 - 8x + 52 \\ (D^2(x))' &= 2x - 8 = 0 \rightarrow x = 4 \\ D^2(4) &= 4^2 - 8 \cdot 4 + 52 = 36 \\ \rightarrow D &= \sqrt{36} = \boxed{6} \end{aligned}$$



- (b) What is the largest area of a rectangle with sides along the axes and a corner on $f(x)$?

$$\begin{aligned} A(x) &= x \cdot 2\sqrt{12-x} \\ A^2(x) &= x^2 \cdot 4 \cdot (12-x) \\ &= 48x^2 - 4x^3 \\ (A^2(x))' &= 96x - 12x^2 = 0 \\ 12x(8-x) &= 0 \rightarrow \begin{matrix} x=0 \\ x=8 \end{matrix} \\ A(0) &= 0 \\ A(8) &= 8 \cdot 2\sqrt{12-8} = \boxed{32} \end{aligned}$$



B5 - Antiderivatives

(a) Find the formula for one function $g(x)$ that satisfies both of the following conditions:

- $g(\pi) = 1$
- $g'(x) = \sec(x) \tan(x)$

$$g(x) = \sec(x) + C$$

$$g(\pi) = \sec(\pi) + C = 1$$

$$-1 + C = 1 \rightarrow C = 2$$

$$\boxed{g(x) = \sec(x) + 2}$$

(b) Find the formula for one function $f(x)$ that satisfies all three of the following conditions:

- $f(1) = 4$
- $f(3) - f(2) = 3$
- $f''(x) = 12x^{-3} + 2$

$$f'(x) = \frac{12 \cdot x^{-2}}{-2} + 2x + C = -6x^{-2} + 2x + C$$

$$f(x) = \frac{-6x^{-1}}{-1} + \frac{2x^2}{2} + Cx + D = 6x^{-1} + x^2 + Cx + D$$

$$f(3) - f(2) = \left(\frac{6}{3} + 3^2 + 3C + D \right) - \left(\frac{6}{2} + 2^2 + 2C + D \right) = 3$$

$$\Rightarrow 11 + 3C + D - 7 - 2C - D = 3$$

$$4 + C = 3 \rightarrow C = -1$$

$$f(x) = \frac{6}{x} + x^2 - x + D$$

$$f(1) = \frac{6}{1} + 1^2 - 1 + D = 4$$

$$6 + D = 4 \rightarrow D = -2$$

$$\boxed{f(x) = \frac{6}{x} + x^2 - x - 2}$$